Project 2

Start Assignment

- Due 1 Apr by 22:00
- Points 10
- Submitting a file upload
- File types html
- Available 18 Mar at 22:00 - 6 Apr at 22:00

> This project draws on lecture and lab content from Module 2: Creating web-based dynamic reporting systems. The knowledge and skills assessed by the project will be covered in lectures/labs by **the end of Module 2**. Consequently, the order of the project instructions below does not necessarily reflect the order the material is covered in lectures/labs. Projects are Lane 2 assessments. More information on **Tirohanga Whānui Aromatawai | Assessments Overview (https://canvas.auckland.ac.nz/courses/141163/pages/tirohanga-whanui-aromatawai-%7C-assessments-overview)** .

Prep work

- You will also need access to a computer that has **R (https://cran.r-project.org/)** installed and **RStudio (https://posit.co/download/rstudio-desktop/)** installed.
- **If installing R and RStudio onto your computer is a problem, you can use the free level of Posit Cloud (https://posit.cloud/)**
- You should already have the R package {tidyverse} installed from Project 1

> Remember, you can use the lab sessions to get help with completing your project, and with any technical difficulties you face, like installing R or RStudio, or any packages.

Recommended project timeline

- Complete Parts A, B, and C after the lectures and lab task for 2A
- Complete Parts D & E after the lectures and lab task for 2B
- Complete Part F after you have completed Parts A, B, C, D, & E

> Remember that you can print this page to PDF, and then use an annotation tool to keep track of your work on the project (e.g., ticking off aspects as you complete them).

Group design and data collection (optional)

Experience working in teams is valuable preparation for your capstone course, can enhance employability and give you better examples to talk about in interviews.

You may choose to work with up to four others students (max group size of five) to complete parts A & B. While you may discuss them with peers/the teaching team, Parts C, D & E must be your own work. Your final submission is individual.

Resource: **Learning Essentials tips (https://learningessentials.auckland.ac.nz/learning-at-university/working-in-groups/)**

Part A: Form design

- The data context for Project 2 is **observational logging**. You need to develop a form/survey that could be completed whenever something is seen or done (i.e., observed).
- You need to select a focus for your data collection. You need to log data based on what **you observe**, for example: on your bus trips, on your daily walks, your Instagram feed, the ads that you get targeted with, the YouTube videos you watch, the clothes people wear, etc.
- 🔑 The key considerations: It has to be something you can observe **multiple** times over a period of roughly a fortnight, AND learn something from analysing the data collected.
- Design questions for your data collection. Think carefully about what aspect(s) of what is observed that you **specifically want to log data about**, and that would be meaningful to track and compare over multiple observations.
- Develop a new Google Form with at least three questions that align with your focus and the following criteria:
    - Uses the four guidelines for designing forms discussed in Module 2, where appropriate or relevant.
    - At least one question that will generate numeric data using a short answer question with response validation.
    - At least one question that will generate categorical data using multiple choice.
    - Appropriate text, visual, and structural elements as needed.
    - Responses are anonymous.
- Change the settings of your Google Form so that it is **NOT** restricted to only users from the University of Auckland. [Note: You can use your non-university Gmail account for this project, in which case this instruction won't apply!]

Part B: Data collection

- Publish the Google Form and change the settings for **responders** so that "Anyone with the link" can respond (under *Manage,* then *Responder view*).
- Generate a Responder link to share the Google Form **with yourself**.
    - It will make your data collection easier if you save this link on the home screen of your mobile phone, or bookmark it on your web browser, so you can access it whenever **you** observe the thing you are focusing on for your observational logging.

- Use your Google form to collect data over several days, ideally at least a week.
  - In total, **you** need to log data at least 20 observations of "thing" you are focusing on, so you have at least 20 rows of data.
- Link your Google Form to a new Google sheet.
- Publish your Google Sheet as a CSV and record this URL (this is a different process from sharing the link or just copying the link to your Google Sheet — see Lecture 2A.2 for more information).
- Keep your form open for collect responses when you submit your project, as the marker will complete your form as part of marking your project.

Part C: Importing data into R

- Within your ***stats220*** folder on your computer, create a new project/folder using RStudio that is called "Project2".
- Create a new R script/file called "exploration.R". It is highly encouraged that you spend time exploring your data first, before you create your dynamic and project reports (Parts E & F).
- Write R code to load the package {tidyverse}.

> Remember, that the package {tidyverse} includes the packages {readr}, {dplyr} and {ggplot2}, so you **don't** need to load these packages as well.

- Use the function `read_csv()` to read data directly from the URL of your published CSV file into a data frame called `logged_data`.

Part D: Exploring data in R

- Keep working in your *exploration.R* file.
- Rename the variables of the data frame `logged_data` using the `rename()` function, as part of creating a new data frame named `latest_data`.
- Use relevant R functions covered in Module 2 to explore your data, with the purpose of identifying at least **two** summary values (e.g., min, max, mean, length, etc.) and at least **two** bar charts that you can use for your dynamic report.

> You need to spend a bit of time getting to know your data and figuring out what key aspects you want to present for your analytics section of the report.
>
> The exploration part of your code should show that you tried different summary values and bar charts as part of analysing your data.

- Ensure your R code uses comments, indenting, and "white space".
- Ensure that your bar charts are informative (see Lab task 2B).

- At the end of your code, you should have a commented section that presents the R code you have decided to use for your report, that produces at least two summary values and at least two bar charts.

Part E: Creating a dynamic report

- Download this ***dynamic_report.Rmd* RMarkdown file (https://canvas.auckland.ac.nz/courses/141163/files/16888146?verifier=ZdBAxbd2lB5PYMXW1weg6Zrxqxap971hRz4Ms1bq&wrap=1) (https://canvas.auckland.ac.nz/courses/141163/files/16888146/download?verifier=ZdBAxbd2lB5PYMXW1weg6Zrxqxap971hRz4Ms1bq&download_frd=1)** and save/move it to your *Project2* folder.
  - Do not change the YAML.
  - The output is purposefully `html_fragment`, do not change this.
  - The title given in the YAML will not be included in the HTML generated, and this is intended.
- Change the level 2 heading to better describe/capture what your dynamic report is tracking.
- Add other markdown text, where appropriate, to communicate the nature of your data and what you are analysing.
- Copy and use relevant R code from your *exploration.R* file to create your dynamic report.
  - You can add more code chunks to your *dynamic_report.Rmd* file if you need.
  - For this report, keep the settings so that `echo=FALSE`, so that none of the code used is printed/visible in the HTML generated.
  - Tidyverse can be loaded "quietly" with `suppressPackageStartupMessages(library(tidyverse))` when you load the library.
- Your dynamic report needs to display at least two bar charts and include “dynamic comments” that are based on at least two summary values.
  - You can use `paste()` or `paste0()` for the dynamic comments, or use inline R code.
- Add any CSS you would like to use for your dynamic report.
- Knit your *dynamic_report.Rmd* file to HTML to check what your dynamic report looks like.
  - You will not submit this HTML file, this is just to support your development of your dynamic report.
- Follow the steps shown in the video below to set up GitHub Pages for your *stats220* GitHub repository, which includes creating new files within your *stats220* repository.
- You will need the following code when creating the *index.html* file within your GitHub *stats220* repo.
  - Do not modify this code.

```
<!DOCTYPE html>
<html>
<head>
<meta http-equiv="Cache-Control" content="no-cache, no-store, must-revalidate">
<meta http-equiv="Pragma" content="no-cache">
<meta http-equiv="Expires" content="0">
<title>Dynamic report</title>
<script src="https://code.jquery.com/jquery-1.10.2.min.js"></script>
<script src="https://cdn.opencpu.org/opencpu-0.4.js"></script>
<script src="https://stat.auckland.ac.nz/~fergusson/web_rmd/web_rmd.js"></script>
</head>
<body style='margin:0'>
<div id="output"></div>
</body>
</html>
```

单击以登录并播放视频

- Record the URL for your GitHub Pages published dynamic report (see video above).
  - The URL should look something like this `https://statscatcode.github.io/stats220/`
- After your dynamic report has been published as a webpage using GitHub pages, check the dynamic nature of the report by using your Google Form to collect more data.

Part F: Writing a project report

We do not recommend use of any type of text generation AI tools for writing your project report. The purpose of the report is for you to report on your learning and experiences completing the project.

- Create a new Rmd file within your *Project2* folder called “project2_report.Rmd”.
- Make the title “Project 2” and put your name as the author.
- Edit the YAML of the *project2_report.Rmd* file so that:
  - the subtitle is “STATS 220 Semester One, 2026”.
  - the code is folded.

- Structure the report (the *project2_report.Rmd* file) using second-level headings as follows:
  - *Introduction*
  - *Dynamic report*
  - *Creativity*
  - *Learning reflection*
  - *Appendix*
- Under the *Introduction* section of your report:
  - Describe the focus for your data collection, including what it was that you observed multiple times over a period of time, e.g., Instagram ads.
  - Discuss one of the guidelines for designing forms and how you considered this when designing your Form.
  - Explain how the data collected would allow you to learn something from analysing the data collected.
  - Use Markdown to include the link to your Google Form (the same one that you shared with yourself — this needs to be accessible to the marker).
  - Group work:
    - If you worked as a group on Parts A 7 B, include the names of your group members: "*Form design and data collection for this project were completed as a group. My group members were ....*".
    - If you did not work in a group, put "*Form design and data collection for this project were completed by me."*
  - Write at least 200 words for this section.
- Under the *Dynamic report* section of your report:
  - Use Markdown to include the link to your GitHub Pages published dynamic report.
- Under the *Creativity* section of your report:
  - Describe and justify how your project demonstrates creativity.

> What does “demonstrate creativity” mean? It means that you have gone beyond what was asked in terms of either your explanations, creations, presentation or use of data technologies, and that the extra things you have done have actually made a better product. Creativity is NOT simply that you have done more than “what was taught”.
>
> For this project, that could mean choosing a focus for your data collection that meets the requirements for the project but was not one of the examples provided in the instructions, re-using skills and knowledge covered in Module 1 **that was not asked for** in this project (e.g., using {magick} functions to create additional relevant images/GIFs for your dynamic report), using regex or other features of Google Forms above what was asked for when designing your form, or producing an **dynamic report** that is statistically insightful and contextually informative.
>
> It **does not** mean just copying and using code from another source that has not been taught in STATS 220 (course, webpage, generative AI tool, etc.). In particular, advanced use of R, including

> functions not taught from the {ggplot2} or {dplyr} packages, will not be recognised as creativity for this project.

- Under the *Learning reflection* section of your report:
  - Describe in your own words at least ONE important idea you learned from Module 2 *Creating web-based dynamic reporting systems*.
  - Discuss what things related to data technologies you are curious about exploring further.
  - Comment on group work
    - If you DIDN'T work in a group for this project, briefly describe one other way you have already, or will in future, develop your group work skills. (Aim for one to two sentences.)
    - If you DID work in a group for this project, describe one thing you learned about successful/effective collaboration. (This could be based on things that went well or things that didn't go so well.) (Aim for one to two sentences.)
  - Write at least 100 words for this section of your report.
- Under the *Appendix* section of your report:
  - Copy the code below exactly as it is given.
  - This will ensure that the contents of your *exploration.R* and *dynamic_report.Rmd* files are included in your project report.

```
```{r file='exploration.R', eval=FALSE, echo=TRUE}
```
```{r file='dynamic_report.Rmd', eval=FALSE, echo=TRUE}
```
```

- Knit your *project2_report.Rmd* file to create a self-contained *project2_report.html* file.

What do you need to submit?

For this project, you will submit just the `project2_report.html` file.

Project 2

Criteria	Ratings	Pts
Design a form for data collection | Form contains required questions and demonstrates use of relevant guidelines for its design		1 pts
Consider data with respect to analysis | Focus for data collection described and explanation given of how data collected would be analysed		1 pts
Store rectangular (tidy) data persistently using Google sheets | Google sheet published as CSV and URL used within analysis		1 pts
Develop R-coding related knowledge | The exploration.R file show use of relevant R functions for reading and renaming data		1 pts
Manipulate tidy data with R | The exploration.R file show use of relevant R functions for summarising data and producing summary values		1 pts
Develop simple graphics with {ggplot2} | The exploration.R file show use of relevant R functions to create and display at least two informative bar charts		1 pts
Create dynamic reports using R Markdown | The dynamic_report.Rmd file contains use of R chunks and markdown that produce bar charts and dynamic comments		1 pts
Develop communication with data and technology | The dynamic report is published using GitHub pages		1 pts
Develop creativity with data and technology | The project demonstrates creativity		1 pts
The project report meets the stated requirements and all correct files were submitted for the project		1 pts
		Total points: 10